

◆ ◆ ◆ T O P S T E C H N O L O G I E S ◆ ◆ ◆

COMPTIA N+

M2-A1 // NETWORK SECURITY, MAINTENANCE
& TROUBLESHOOTING PROCEDURES

■ ■ ■ MERGED QUESTIONS + ANSWERS DOSSIER ■ ■ ■

SYSTEM SPECIFICATIONS

PROGRAM **CompTIA N+** CODE **M2-A1**

DURATION **3 Hours** STACK **Networking Hardware**

Sections A–D // Concept · Practical · Capstone · AI-Augmented

A

SECTION A

CONCEPT APPLICATION

S1

◆ SCENARIO

Nexus Innovations is purchasing new computers for employees who use programming software, web browsers, and video conferencing applications at the same time.

► QUESTION

What factors should be considered when selecting the CPU, RAM, and storage to ensure good performance and hardware compatibility?

◆ ANSWER

The core constraint here is concurrent multi-process load, not raw single-task horsepower, so selection criteria should center on sustained multitasking rather than peak burst speed:

- **CPU:** Prioritize core/thread count over raw clock speed — a 6–8 core CPU with hyperthreading handles an IDE compiling code, a browser with many open tabs, and video-conferencing encode/decode simultaneously far better than a high-clock 4-core chip. Check socket compatibility with the target motherboard and TDP against the case's cooling capacity. A security-conscious build also avoids CPUs without VT-x/AMD-V virtualization support, since dev workflows (Docker, VMs, sandboxed testing) and antivirus sandboxing both depend on it.
- **RAM:** 16GB minimum, 32GB preferred — IDEs, containers, and Electron-based conferencing apps are memory-hungry. Confirm the motherboard's supported RAM speed/generation (DDR4 vs DDR5) and run in dual-channel for bandwidth.
- **Storage:** NVMe SSD, not SATA SSD or HDD — compilation, package installs, and OS swapping are I/O-bound. Also factor in hardware-accelerated encryption support (AES for BitLocker/FileVault), since dev machines often hold source code and credentials worth protecting at rest.
- **Compatibility:** Confirm CPU socket ↔ motherboard chipset, RAM generation ↔ motherboard support, and PSU wattage headroom for the CPU/GPU combination chosen.

S2

◆ SCENARIO

After installing antivirus software on all startup office computers, employees notice that their systems have become slower.

► QUESTION

Explain how the antivirus software affects CPU, RAM, and storage performance, leading to system slowdowns.

◆ ANSWER

Antivirus introduces overhead across all three subsystems because real-time protection means every file operation is intercepted and inspected before it completes:

- **CPU:** Real-time scanning hooks into file I/O and process execution, running signature-matching and heuristic/behavioral analysis on every opened file or launched process. This consumes cycles that would otherwise go to user applications, especially during full-system or scheduled scans.
- **RAM:** The AV engine, its signature database, and behavioral-monitoring cache stay resident in memory; some suites also run isolated sandbox processes for suspicious files, adding further memory pressure.
- **Storage:** Every read/write is intercepted for on-access scanning, adding latency to disk operations — most noticeable on HDDs or when scanning large project directories with thousands of small files.

The fix isn't disabling protection — it's tuning: exclude trusted directories from real-time scanning, schedule full scans off-hours, and confirm the AV isn't duplicating protection already built into the OS (running two real-time engines simultaneously compounds overhead and can even create conflicts that mimic malware behavior).

S3

◆ SCENARIO

A USB keyboard and printer work properly on one computer but do not work correctly on another computer in the startup office.

► QUESTION

Describe the steps you would follow to troubleshoot the problem if it is caused by driver incompatibility or operating system issues.

◆ ANSWER

1. Isolate the variable — swap the same USB cable/port between machines to rule out a bad port or cable before blaming drivers.
2. Check Device Manager (Windows) for a yellow warning icon on the device — this immediately confirms whether it's a driver problem vs. a power/enumeration problem.
3. Verify driver status — outdated, corrupted, or missing driver; try "Update driver" first, then a clean uninstall + reboot (Windows will often re-detect and reinstall a working driver).
4. Check OS/version compatibility — an older driver package may not support a newer OS build, or vice versa; confirm against the vendor's supported OS list.
5. Check USB power management — Windows can selectively suspend USB ports to save power, which sometimes drops peripherals; disable "USB selective suspend" as a test.
6. **Security angle** — confirm no Group Policy or endpoint-security tool (USB device control policy) is blocking the device. In an office environment, a security tool restricting unauthorized USB devices can present identically to a driver failure, and this is often the overlooked step.
7. Firmware/BIOS — as a last resort, check BIOS USB settings (legacy USB support enabled) and update firmware if available.

S4

◆ SCENARIO

Nexus Innovations experiences frequent power cuts and voltage fluctuations.

► QUESTION

Explain how a UPS and surge protector help protect computer hardware and prevent data loss during these situations.

◆ ANSWER

These solve two different failure modes:

- **Surge protector:** Passively clamps voltage spikes (lightning, grid switching) above a threshold and diverts the excess to ground, protecting components from instant hardware damage. It does nothing during an actual outage or brownout.
- **UPS (Uninterruptible Power Supply):** Provides battery-backed power that bridges the gap during an outage, giving enough runtime to save open work and perform a graceful shutdown — this is what actually prevents data loss and filesystem corruption from an abrupt power cut. A line-interactive or double-conversion UPS also actively regulates voltage during brownouts/sags, which a surge protector cannot do.

Together they form a layered defense: the surge protector guards against destructive spikes, and the UPS guards against interruption and enables a controlled shutdown — conceptually the same defense-in-depth logic used in network security, where multiple layers each cover a gap the others leave.

S5

◆ SCENARIO

New employees at Nexus Innovations report that they can reach internal systems (e.g. the application server) by IP address but not by name, and their devices sometimes fail to obtain a valid IP address when they connect to the network.

► QUESTION

Explain how DNS and DHCP work together in this network, and identify which service is most likely misconfigured based on the symptoms described.

◆ ANSWER

- **DHCP** dynamically assigns an IP address, subnet mask, default gateway, and (critically) DNS server addresses to a client when it joins the network.
- **DNS** translates hostnames (e.g., appserver.nexus.local) into IP addresses so users don't need to memorize numeric addresses.

Symptom analysis

- **"Reach systems by IP but not by name"** → classic DNS failure symptom. Either the DHCP-assigned DNS server address is wrong/unreachable, the DNS server itself isn't resolving internal records, or the client's DNS cache is stale.

- **"Devices sometimes fail to get a valid IP"** → DHCP failure symptom — likely causes include DHCP scope exhaustion (pool ran out of leasable addresses), a rogue/duplicate DHCP server on the network handing out conflicting addresses, or the DHCP service itself being down or misconfigured.

Most likely misconfiguration

Given both symptoms appear together, this points to the DHCP server handing out an incorrect or unreachable DNS server address in its scope options — clients still get an IP (so DHCP is partially working) but inherit bad DNS settings, while the intermittent IP failures suggest the DHCP scope itself may be too small or misconfigured for the network size. Fix: verify the DHCP scope size and lease duration, confirm the DNS server IP configured in DHCP options matches a functioning internal DNS server, and check for a second unauthorized DHCP server on the LAN — a rogue DHCP server is also a known attack vector, since man-in-the-middle setups often exploit exactly this weakness.

S6

◆ SCENARIO

After the IT team installs a new firewall at Nexus Innovations, employees can access the internet but are unable to reach the critical internal application server.

► QUESTION

Describe the troubleshooting steps you would take to determine whether the firewall rules are blocking legitimate traffic, and how you would resolve the issue.

✦ ANSWER

8. Confirm the symptom scope — internet access works (so the default route/NAT/outbound rules are fine); the problem is isolated to inbound/internal rule logic, likely a new deny-by-default policy blocking traffic that wasn't explicitly permitted.
9. Check the rule base top-down — firewall rules are typically processed in order with first-match-wins. A broad "deny all" rule placed above the specific "allow to app server" rule will silently block legitimate traffic even if the allow rule exists further down.
10. Verify the specific rule exists and is correct — source zone/IP, destination IP/port, and protocol (TCP/UDP) must exactly match the application server's actual listening port and the clients' subnet.
11. Check NAT/zone assignments — confirm the app server's interface and the client segment are in zones the firewall is actually configured to permit traffic between. A common post-install mistake is leaving new zones in a default-deny state.
12. Test methodically — use ping, telnet <ip> <port>, or Test-NetConnection to isolate whether it's a full block (ICMP and TCP both fail) or a Layer-4 rule issue (ping succeeds, port connection fails).
13. Check logs — firewall deny logs will show exactly which rule is dropping the traffic; this is the fastest path to root cause rather than guessing.
14. Resolve — add or reorder an explicit allow rule for the required source→destination→port, then retest and document the change, applying least-privilege by allowing only the specific port/service needed, not a blanket allow.

Complete all tasks using: PC Tool Kit, Cisco Packet Tracer, Windows/Linux ISO, and VirtualBox. These are hands-on, photo-documented lab tasks — the guidance below outlines the correct procedure and validation criteria for each.

B

SECTION B

PRACTICAL APPLICATION

Task 1 – Workstation Assembly

◆ TASK

Physically assemble a workstation for the startup office and document the component installation sequence. Install CPU, RAM, storage, and other core components in the correct sequence; connect all required internal cables (power, data, front panel); document each step; system must pass POST after assembly.

◆ ANSWER

1. Ground yourself (anti-static wrist strap / touch the case chassis) — ESD is the #1 preventable failure cause.
2. Install the CPU into the socket (align the notch/pin-1 indicator), apply thermal paste, and mount the cooler.
3. Install RAM into the correct DIMM slots (check the motherboard manual for dual-channel slot pairing — usually slots 2 and 4).
4. Mount the motherboard into the case on standoffs.
5. Install the PSU, then storage drives (NVMe into the M.2 slot, or SATA SSD/HDD into drive bays).
6. Connect front panel headers (power switch, reset, LED, USB, audio) — this is the step most people get wrong; follow the motherboard manual's pinout diagram exactly.
7. Connect the 24-pin motherboard power, 4/8-pin CPU power, and SATA/PCIe power to drives.
8. Install the GPU if applicable and connect its power.
9. Cable-manage, close the case, and connect peripherals plus the monitor.
10. Power on and confirm POST (beep code or POST LED / splash screen) before installing any OS.

Document each step with a labeled photo; a successful POST (BIOS splash screen appears, no beep-code error) is the proof of successful assembly.

Task 2 – RAM Mismatch Troubleshooting

◆ TASK

Troubleshoot and resolve a RAM mismatch error within BIOS/UEFI: identify the error via POST messages, determine the root cause, resolve it, and confirm the system boots normally.

◆ ANSWER

- A POST error or beep-code pattern indicates a RAM fault — check the motherboard manual for the specific beep-code meaning.
- Boot into BIOS/UEFI and check the memory info page for the detected speed/capacity per slot versus the rated spec.

- Common root causes: mixed RAM speeds (the system will run at the slowest module's speed, or fail to boot if XMP/timings conflict), mismatched capacity in a dual-channel configuration, or a module that isn't fully seated.
- Fix: reseat modules fully (listen/feel for the click), match RAM in identical slot pairs, apply the correct XMP/DOCP profile in BIOS if using non-JEDEC speeds, and retest one module at a time if the issue persists, to isolate a physically faulty stick.

Task 3 – LAN Topology Build

◆ TASK

Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet; all assigned IPs must be reachable via Ping.

✦ ANSWER

- A default Class C range such as 192.168.1.0/24 provides 254 usable hosts — far more than needed for 5 PCs + 1 printer. Carving a smaller block, e.g. 192.168.1.0/29 (6 usable hosts), exactly fits 5 systems + 1 printer with zero waste and pairs directly with the subnetting skill exercised in Section D.
- Topology: a switch connecting 5 PCs + 1 network printer; a router is optional if only testing local connectivity.
- Assign static IPs within the chosen subnet, matching the subnet mask and default gateway on all devices.
- Validate with ping from each PC to every other device (including the printer) — all pings must succeed before submission.

Task 4 – Clean OS Installation

◆ TASK

Perform a clean OS install with manual disk partitioning (not guided) and startup optimization; OS must boot without errors.

✦ ANSWER

- Boot from the ISO (USB or virtual media in VirtualBox) and select the "Custom/manual" install option — never "guided."
- Manually create partitions: an EFI System Partition (~100–500MB, FAT32), and a primary OS partition (remainder of the drive, NTFS for Windows / ext4 for Linux). For a more resilient build, consider a separate partition for user data to simplify future re-imaging without data loss.
- Complete the install, then in Task Manager (Windows) → Startup tab, or via systemctl (Linux), disable unnecessary startup programs/services to reduce boot time and shrink the system's attack surface — fewer auto-starting services means fewer background processes that could be exploited.
- Confirm a clean boot to desktop/login with no error dialogs.

Nexus Innovations – Diagnosing Random System Reboots

◆ OBJECTIVE

At Nexus Innovations, three rendering workstations pass POST every morning but restart randomly during heavy afternoon workloads. The computers are kept inside poorly ventilated desk cabinets. Analyze the probable hardware and thermal causes and propose a remediation plan, combining hardware, power, and troubleshooting concepts.

✦ ANSWER

Three Most Likely Hardware Culprits Under Sustained Afternoon Load

1. **PSU (Power Supply Unit)** — if undersized or degrading, it can't sustain peak wattage draw under sustained GPU/CPU load, causing a protective shutdown/restart.
2. **CPU/GPU thermal throttling leading to shutdown** — poor ventilation in a closed cabinet traps heat; once components hit critical temperature, thermal protection forces a hard restart to prevent damage.
3. **RAM instability under load** — marginal or overheating RAM can pass POST (a low-load, momentary state) but fail under the heavier memory access patterns of rendering workloads, causing crashes/reboots.

Why the Systems Pass POST in the Morning but Fail in the Afternoon

POST is a low-load, momentary diagnostic — it doesn't stress components anywhere near their thermal or power limits. The cabinet's poor ventilation means heat accumulates progressively through the day; by the afternoon, ambient case temperature is already elevated from earlier hours of rendering work, so the same workload now pushes components past their thermal threshold. It's a cumulative thermal load issue rather than a static hardware failure, which is exactly why the symptom is intermittent rather than constant.

Using a Stress-Test Tool to Confirm a Thermal or Power Issue

Run a combined stress tool (e.g., Prime95 for CPU, FurMark or 3DMark's stress test for GPU, or a combined tool like AIDA64/OCCT that stresses CPU, RAM, and PSU simultaneously) while monitoring real-time temperatures and voltages with HWMonitor or the motherboard's own sensor software. Reproduce the failure under controlled, monitored conditions: if temperatures climb steadily toward the component's thermal limit right before a crash, or voltage rails sag/fluctuate under load, that confirms a thermal or power delivery root cause rather than a software or driver issue.

Three Remediation Steps

1. **Improve airflow/ventilation** — relocate workstations out of the closed cabinets, or add active cabinet ventilation (exhaust fan, vented panels) so hot air has somewhere to go.
2. **Improve case-level cooling** — add or upgrade case fans for better intake/exhaust airflow, reapply thermal paste on the CPU (and GPU if applicable) on older machines, and confirm heatsinks are properly seated.
3. **Implement thermal monitoring/alerting** — deploy lightweight monitoring (e.g., HWiNFO with logging, or an IT asset monitoring agent) so temperature trends are visible before they cause another failure. Proactive detection rather than reactive troubleshooting is a habit worth carrying into general IT operations.

Is a PSU Upgrade Necessary? — Technical Justification

A PSU upgrade is justified if the stress test shows voltage sag/instability under load, or the PSU's rated wattage is genuinely undersized for the CPU+GPU combination doing render work. However, if the stress test shows stable voltages but climbing temperatures, the PSU is not the root cause — replacing it would treat a symptom while leaving the actual thermal/ventilation problem unresolved. The stress-test data should drive the decision, not assumption.

D

SECTION D

AI-AUGMENTED LEARNING

Step 1 • Build with AI

Use an AI tool of your choice to generate an IP addressing and subnetting plan for Nexus Innovations' startup office network, covering 5 systems and 1 printer. The plan must specify the subnet, subnet mask, usable host range, and broadcast address; allocate addresses for all 5 systems and 1 printer; identify addresses to reserve for future expansion; and document the exact prompt(s) used.

Step 2 • Test & Debug (without AI)

Verify the plan for errors — overlapping ranges, incorrect network/broadcast addresses, insufficient host capacity, or unreserved growth space — and correct them yourself.

SUBMIT — ALL 3 ITEMS REQUIRED

1. The exact prompt(s) given to the AI tool.
2. The AI's original subnetting plan and the corrected version.
3. A 3–4 line note explaining what was changed and why the AI's version needed correction.

⚠ STATUS

This section requires the student's own individual AI interaction and documented, unaided review — it is intentionally left for the student to complete using their own AI prompt, per the assessment's instructions.

■ ■ ■ END OF assignment ■ ■ ■